

DTU NCC

Official Cadet Portal — Website Specification

Attendance & Fine Management • Inter-Squadron Competitions & Leaderboard • Camp Information • Cadet Login & Role-Based Access

Prepared for	CQMS, DTU NCC
Squadrons Covered	Alpha • Bravo • Charlie • Delta
Document Type	Functional & Technical Specification
Purpose	Blueprint for design and development of the DTU NCC website

This document lays out the complete structure, features, user roles, database design, and recommended technology stack for the proposed DTU NCC website.

1. Project Overview

The DTU NCC website will serve as the single digital home for the unit — combining cadet records, attendance tracking with an automatic fine system, inter-squadron competitions with a live leaderboard, and camp information. It will have two broad zones: a **public zone** (visible to anyone, showcasing DTU NCC) and a **cadet/management zone** (accessible only after login, showing personal and administrative data).

- **Goal 1 — Attendance & Discipline:** Make attendance compulsory and transparent; automatically fine ■50 per absence and let cadets track their own dues.
- **Goal 2 — Competitions:** Run inter-squadron competitions (Alpha, Bravo, Charlie, Delta) with a public, always-updated leaderboard.
- **Goal 3 — Camps & Info:** Publish camp schedules, eligibility, and circulars in one place instead of scattered WhatsApp messages.
- **Goal 4 — Access Control:** Give cadets a personal login, and give CQMS/officers a management dashboard, with clear boundaries on who can edit what.

2. Website Structure (Sitemap)

A. Public Pages (no login needed)

- **Home** — Hero banner, About DTU NCC (motto, wings, history), highlights, quick links.
- **About Us** — NCC intro, DTU unit history, Wing details (Army/Navy/Air, whichever applies), officer list (ANO/CTO), achievements gallery.
- **Competitions & Leaderboard** — Live squadron standings (Alpha/Bravo/Charlie/Delta) and a list of past/upcoming competitions — visible to everyone for transparency.
- **Camps** — Upcoming and past camps (e.g. CATC, ATC, RDC/TSC-type camps), dates, venue, eligibility.
- **Gallery / Achievements** — Photos and cadet/unit achievements.
- **Contact / Notice Board** — Announcements, contact details.
- **Cadet Login** — Entry point into the private zone.

B. Private Zone — Cadet Dashboard (after login)

- **My Profile** — Name, enrollment/regimental number, squadron, rank, wing, contact info, photo.
- **My Attendance** — Calendar/list view of present/absent days, running attendance percentage.
- **My Fines** — Auto-generated ■50-per-absence ledger, total due, paid/unpaid status, history.
- **My Achievements** — Certificates, camps attended, competitions participated in, badges/medals.
- **My Squadron** — Squadron's current leaderboard rank and points breakdown.
- **Notifications** — Camp alerts, parade timing changes, fine reminders, competition schedules.

C. Private Zone — Management Dashboard (CQMS / Officers / Admin)

- **Cadet Database** — Add/edit/remove cadets, assign squadron, ranks, wing.
- **Attendance Marking** — Mark daily/parade attendance per cadet or squadron; bulk-upload option.
- **Fine Management** — Auto-calculated fines, ability to waive/edit a fine (with reason logged), mark fines as paid, export due list.
- **Competitions Manager** — Create a competition, assign points per squadron per event, auto-recalculate leaderboard.
- **Camp Manager** — Add camp details, upload circulars/PDFs, publish selected-cadet lists.
- **Announcements** — Post notices to homepage/notification feed.
- **Reports** — Attendance %, fines collected, squadron performance — exportable to Excel/PDF.
- **User & Role Management** — Control who has admin rights (see Section 3).

3. User Roles & Access Control

Access is layered so that ordinary cadets can only see their own data, squadron officers can manage their squadron, and the CQMS/ANO have full control. Suggested roles:

Role	Who	Access
Visitor	Anyone on the internet	Home, About, Camps info, Gallery, public Leaderboard — read-only, no login
Cadet	Every enrolled NCC cadet	Own profile, own attendance, own fine ledger, own achievements, squadron rank — read-only for everything except their own basic profile edits (e.g. contact info)
Squadron Officer (e.g. SUO/JUO/CSM of Alpha/Bravo/Charlie/Delta)	Senior cadets heading a squadron	Everything a Cadet has, plus: mark attendance for their own squadron, view their squadron's fine list
CQMS (You) / Admin	Cadet Quarter Master Sergeant	Full management dashboard: all cadets, all squadrons, attendance, fines (including waive/edit), competitions, camps, announcements, reports
ANO / Officer-in-Charge	Associate NCC Officer / Caretaker Officer	Same as CQMS plus final approval on fine waivers and camp nominations, and overall analytics view
Super Admin	Website maintainer/developer (may rotate each year)	Technical access: manage user accounts & roles, backups, site settings — separate from the day-to-day NCC management role

Tip: keep 'Super Admin' (technical) separate from 'CQMS/ANO' (organisational) so that changing a website maintainer each year doesn't require touching NCC records.

4. Attendance & Fine System — How It Should Work

- Attendance is taken per parade/drill day by the Squadron Officer or CQMS, either cadet-by-cadet or via a marked register uploaded in bulk.
- Any cadet marked **Absent** automatically gets a **■50 fine** added to their ledger — no manual fine entry needed.
- A cadet can request an **exemption** (e.g. medical leave) before/after the parade; this needs CQMS/ANO approval and, if approved, does not add a fine but is still logged as 'Excused Absence'.
- Each cadet's dashboard shows: total classes held, attended, absent, attendance %, and current fine balance with a payment-status tag (Paid / Unpaid / Partially Paid).
- CQMS dashboard shows a unit-wide due list, sortable by squadron, so pending fines can be followed up in person or in the next parade.
- Recommended: fines are tracked digitally but **collected physically/manually** (cash or UPI) since a cadet club rarely needs a full payment gateway — the CQMS simply marks a fine 'Paid' once received. A UPI QR code can optionally be shown on the fines page for convenience.

5. Inter-Squadron Competitions & Leaderboard

- CQMS creates a competition (e.g. Drill Competition, Quiz, Sports Meet, Cultural Night) and defines how points are awarded (e.g. 1st = 10 pts, 2nd = 7 pts, 3rd = 5 pts, participation = 2 pts).
- After the event, results are entered once — the website automatically adds the points to each squadron's running total.

- The **Leaderboard** page (public) always shows current standings: Alpha, Bravo, Charlie, Delta ranked by total points, plus a breakdown of which competitions contributed to each squadron's score.
- A competition history log keeps every past event and result on record for future reference.

Squadron	Example Points	Rank
Alpha	142	1st
Charlie	128	2nd
Bravo	119	3rd
Delta	101	4th

Illustrative example of how the public leaderboard table would look.

6. Camps Section

- List of camps (e.g. CATC, ATC, various types of camps your unit attends) with dates, venue, and eligibility criteria.
- Downloadable circulars/instructions (PDF) for each camp.
- Published list of selected/nominated cadets per camp.
- Past-camps archive with photos/reports for reference.

7. Suggested Database Design

Below are the core data 'tables' the website needs. This is meant as a guide for whoever builds the backend — each bullet is one table with its key fields.

Table	Key Fields
Cadets	cadet_id, name, enrollment_no, squadron, rank, wing, contact, photo, login credentials
Squadrons	squadron_id, name (Alpha/Bravo/Charlie/Delta), total_points
Attendance	record_id, cadet_id, date, status (Present/Absent/Excused), marked_by
Fines	fine_id, cadet_id, attendance_record_id, amount (default ■50), status (Paid/Unpaid), date_paid
Competitions	competition_id, name, date, description, point_scheme
Competition_Results	result_id, competition_id, squadron_id, position, points_awarded
Camps	camp_id, name, start_date, end_date, venue, eligibility, circular_file
Camp_Nominations	nomination_id, camp_id, cadet_id, status
Users	user_id, cadet_id (nullable), role (Cadet/Squadron Officer/CQMS/ANO/Super Admin), password_hash
Announcements	announcement_id, title, body, posted_by, date

8. Recommended Technology Stack

Two realistic options, depending on how much coding support you have. Both fully support everything above — logins, attendance, fines, leaderboard, camps, and role-based access.

Option A — Fastest to Build (recommended for a cadet-run project)

Layer	Technology
Frontend	HTML, CSS, JavaScript or React.js
Backend + Database	Firebase (Firestore + Firebase Authentication) or Supabase (Postgres + Auth built in)
Hosting	Firebase Hosting / Vercel — both have free tiers, no server to maintain
Why	No backend server to run or pay for; authentication, database, and file storage (for circulars/photos) all come built-in; easiest to hand over to next year's CQMS

Option B — Full Custom Build (if you or a teammate wants to code it end-to-end)

Layer	Technology
Frontend	React.js + Tailwind CSS
Backend	Node.js + Express.js (REST API)
Database	MongoDB (flexible) or PostgreSQL/MySQL (structured, good fit for attendance/fines tables)
Authentication	JWT-based login with bcrypt-hashed passwords, role field per user
File storage	Cloudinary or Firebase Storage for photos and camp circulars
Hosting	Frontend on Vercel/Netlify; backend on Render/Railway; database on MongoDB Atlas or Supabase
Why	Full control over logic (fine rules, leaderboard scoring, reports); good if this is also meant as a learning/portfolio project

Either option supports the same features — the choice is really about how much backend coding you want to take on. Many college-club sites start with Option A and migrate to Option B later if needed.

9. Suggested Build Order (Phases)

Phase	What to Build
1	Public site: Home, About, Camps (static info), Contact — get DTU NCC online first
2	Login system + Cadet Dashboard (profile view only)
3	Attendance module + automatic ■50 fine logic + fine ledger view
4	Competitions module + public Leaderboard
5	Full CQMS/Admin management dashboard + reports/export
6	Polish: notifications, gallery, announcements, mobile responsiveness

This document is intended as a working blueprint — feel free to adjust point values, fine amounts, squadron names, or roles as your unit's requirements evolve.